

JIHO KIM

[Email](#) | (608) 949 2397 | [LinkedIn](#) | [Google Scholar](#) | [GitHub](#)

SUMMARY

Ph.D. student in Computer Sciences studying data visualization and human-computer interaction. Research focuses on visual perception, user modeling, accessibility, and AI-assisted visualization design.

EDUCATION

University of Wisconsin-Madison

Ph.D. in Computer Sciences

May 2027 (Anticipated)

M.S. in Computer Sciences

May 2025

University of Southern California

B.S. in Computer Science

May 2020

GPA: 3.70/4.00, Major GPA: 3.83/4.00

PUBLICATIONS

Peer-reviewed

- **Kim, J.**, Chinni, A., Schloss, K. B., & Gleicher, M. (2026). Divided Attention Amplifies the Importance of Expectation-Aligned Visualization Design. IEEE VIS 2026, forthcoming
- Kim, G. *, & **Kim, J.** *, & Kim, Y. S. (2023) "Explain What a Treemap is": Exploratory Investigation of Strategies for Explaining Unfamiliar Chart to Blind and Low Vision Users. ACM CHI 2023
- **Kim, J.**, Srinivasan, A., Kim, N., Kim, Y. S. (2023) Exploring Chart Question Answering for Blind and Low Vision Users. ACM CHI 2023

Manuscripts in Preparation

- **Kim, J.**, Schloss, K. B., & Gleicher, M. Comparison Axis Alignment: Characterizing the Interaction of Layout, Mark, and Task in Multi-Chart Comparison
- **Kim, J.**, Zhao, Y., Zhu, X., & Kim, Y. S. Machine teaching for accessible line chart sampling

RESEARCH EXPERIENCE

Graduate Researcher

September 2021 - Present

University of Wisconsin-Madison

- Visualization design under divided attention
 - o Designed controlled experiments to measure how divided attention affects visualization perception, mapping, and comprehension.
 - o Modeled user cognitive process with Linear Ballistic Accumulator.
 - o Identified increased reliance on expectation-aligned visual designs under divided attention.
- Reconceptualizing multi-chart layouts - [GitHub](#)
 - o Characterized layout as the joint effect of layout, mark, and task and used this framework to predict viewer performance of comparison across charts.

- Eye tracking for user modeling
 - o Designed eye-tracking studies to characterize strategies people use when interpreting data visualizations.
- Accessible sampling using perceptual optimization - [GitHub Repo](#)
 - o Developed an optimization pipeline to improve the efficiency of analyzing line chart trends for blind users by applying the machine teaching framework.
- LLM visualization design
 - o Evaluated LLM reasoning about visualization design and identified recurring strategies for AI-assisted design.
- Generating text explanations of difficult chart types
 - o Designed strategies for explaining unfamiliar chart types (e.g., treemaps), leveraging learning theories and automatic text generation.
- Chart question answering for the blind and low-vision
 - o Analyzed blind users' interaction with chart question-answering (QA) systems and identified design requirements for a more accessible chart QA system.

Undergraduate Research

January 2018 - May 2018

University of Southern California Information Sciences Institute

- Assisted the development of FRADE, a framework to distinguish DDoS from flash crowd events.
- Managed the simulation and analysis of network traffic in a Linux environment.

TEACHING EXPERIENCE

Teaching Assistant

September 2021 - Present

University of Wisconsin-Madison

- Taught "CS240: Introduction to Discrete Mathematics" and "CS839: Modeling User Interaction"

HONORS & AWARDS

- Student Research Grants Competition Award (\$1,500) May 2023
- UW-Madison CS Department Summer Research Fellowship May 2022
- 6th place, ICPC Southern California Regional, 2018 November 2018
- 6th place, USC Programming Contest, 2018 September 2018

ADDITIONAL EXPERIENCE

Republic of Korea Army Interpreter

September 2015 - June 2017

203rd Special Assault Brigade, 2nd Operational Command

- Coordinated translation between Korean and US military units for critical military operations including the nationwide military drills Key Resolve and Ulchi-Freedom Guardian.

Chart question answering with FeTaQA - [Manuscript](#)

- Implemented and modified the end-to-end model with BART and analyzed its performance.

USC Interaction Lab website - [Website](#)

- Overhauled the website using Firebase and Vue.js. Managed the UX and database design.

StockOverflow - [GitHub Repo](#)

- Developed a gamified stock trading web app using PHP, JavaScript, and MySQL.